Mathematical Modeling of Small Gene Regulatory Networks Reveals Key Regulators and Network Properties Important for Controlling the Early Response to Cold Shock in *Saccharomyces cerevisiae*

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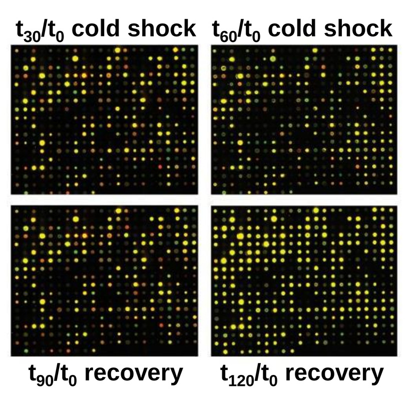
Yeast Respond to Cold Shock by Changing Gene Expression

- The cold shock response can be divided into an early and late response.
- General Environmental Stress Response (ESR) genes are induced in the late response.
- Late response is regulated by the Msn2/Msn4 transcription factors.
- No "canonical" transcription factor responsible for the early response?
  1. Which transcription factors control the early response?
  2. What are their relative levels of influence?
  3. What are the indirect effects of other transcription factors in the *gene regulatory network*?

Microarray Data from Yeast Cold Shock Experiments Were Used to Derive a Family of Related Gene Regulatory Networks



- DNA microarray data from six strains was analyzed to show which genes exhibited significant changes in expression during cold shock (15, 30, 60 min. at 13°C) and subsequent recovery (30, 60 min back at 30°C). The criteria for significance was a Benjamini & Hochberg-corrected ANOVA *p* value < 0.05.

Strain	Wild type	Δ*cin5*	Δ*gln3*	Δ*hap4*	Δ*zap1*
Number genes with significant change in expression at any timepoint	1936 (31%)	1683 (28%)	1683 (28%)	1794 (29%)	1859 (30%)

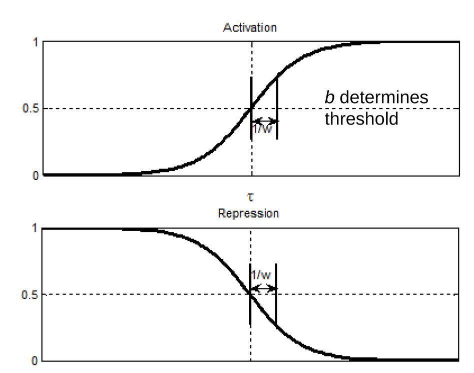
- The genes with significant expression changes were submitted to the YEASTRACT database, which returned a list of transcription factors that could regulate those targets.
- Six related gene regulatory networks with 14-17 nodes and 25-35 edges were then constructed, which included the transcription factors for which deletion strain microarray data were available.

A Nonlinear Differential Equation Models the Expression of Each Transcription Factor in the GRNs Over Time

- GRNmap (Gene Regulatory Network modeling and parameter estimation) was used to model the dynamics of each GRN in response to cold shock (Dahlquist et al. 2015).
- Each transcription factor has a differential equation that models its expression over time.

$$\frac{dx_i(t)}{dt} = \frac{P_i}{1+\exp\left(-\sum_j \left(w_{ij}x_j(t)-b_j\right)\right)} - d_i x_i(t)$$

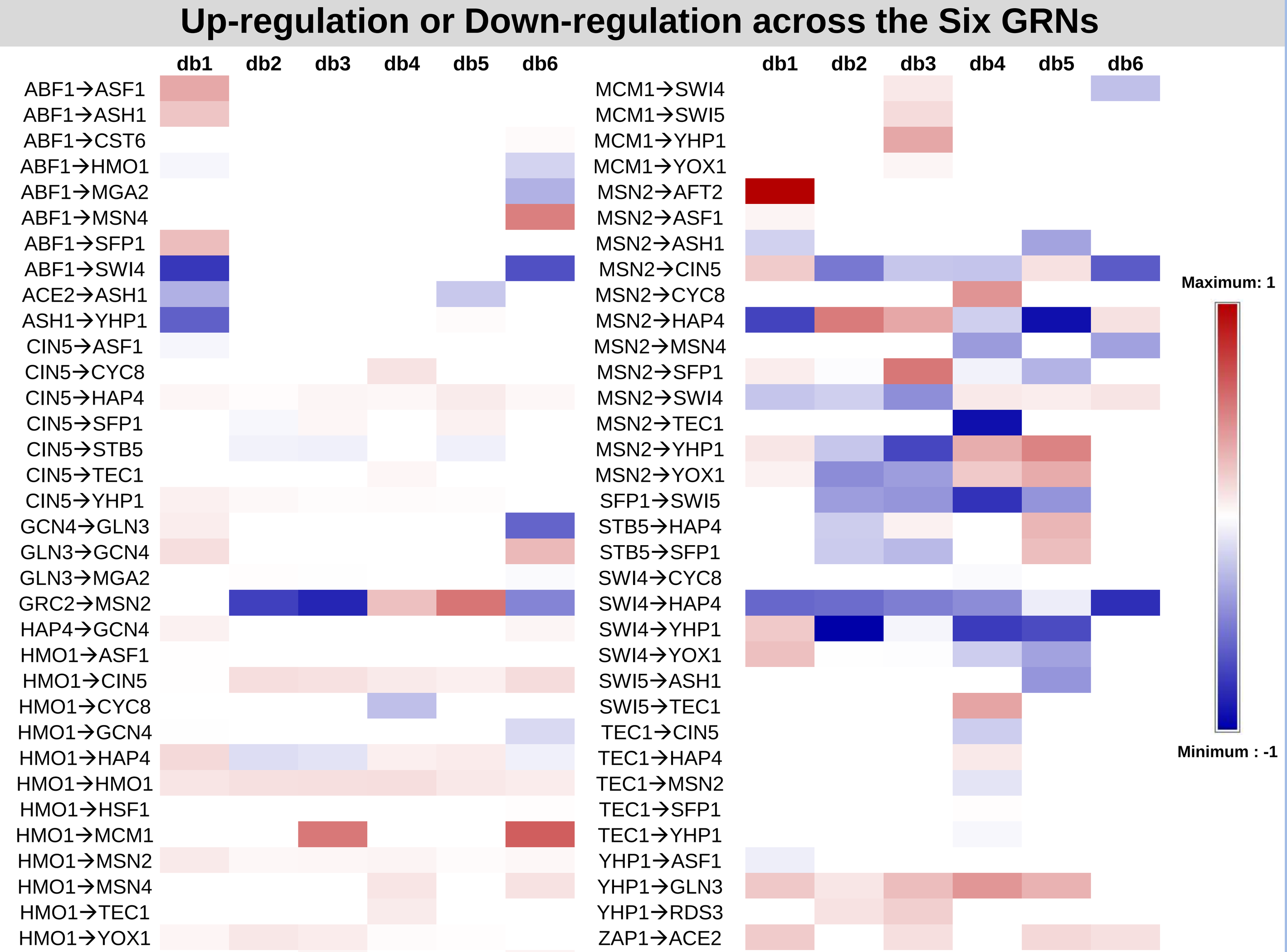
- We use a sigmoidal production function where:
  - P_i is mRNA production rate for gene i
  - d_i is the mRNA degradation rate for gene i
  - w is weight term, determining the level of activation or repression of j on i
  - b is a unique threshold for each gene
- Degradation rates for each gene were derived from half lives reported by Neymotin et al. (2014).
- The production rate (P_i), weight (w), and threshold (b) values were estimated from DNA microarray data using a penalized least squares approach.



$$E = \alpha\|\theta\|^2 + \frac{1}{Q}\sum_{r=1}^{Q}[z^d(t_r) - z^c(t_r)]^2$$

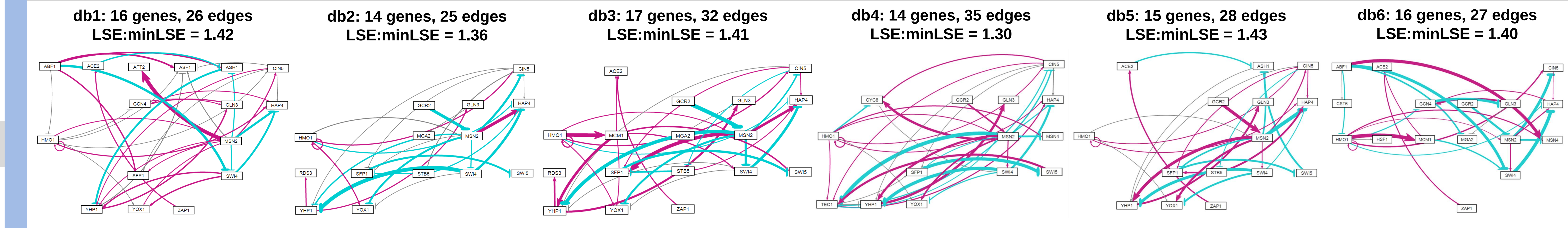
- E represents the least squares error between estimated values and microarray data values.
- θ is the penalty term, which is the combined w, P, and b parameter values.

Estimated Regulatory Weight Terms Generally Show Either Consistent Up-regulation or Down-regulation across the Six GRNs



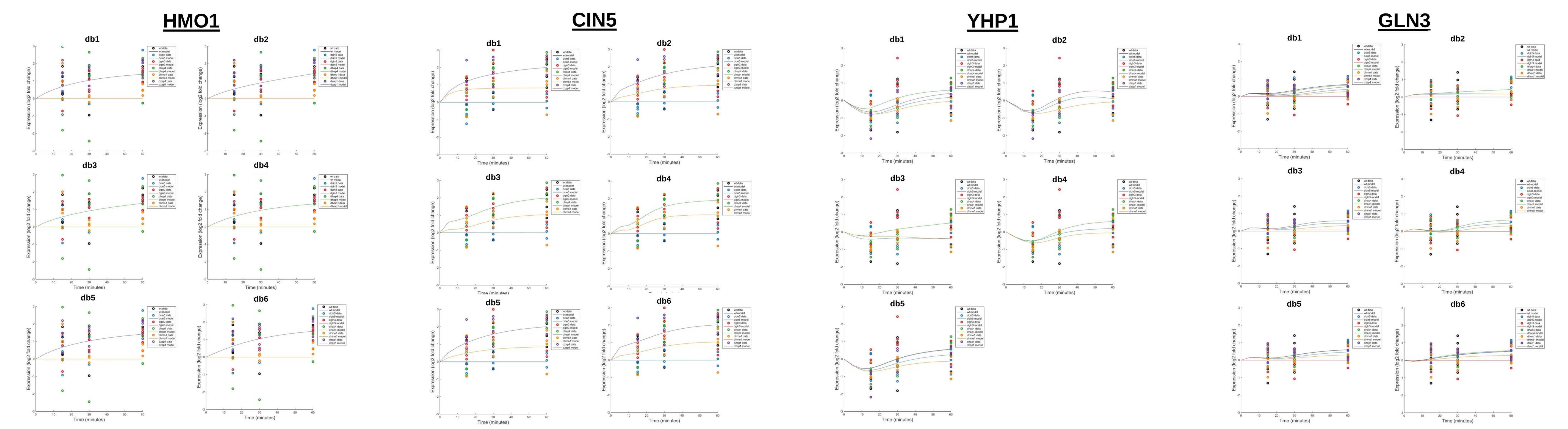
- Regulatory relationships are written as regulator→target; not all relationships were present in all six networks.
- Regulatory weights obtained from GRNmap were normalized to the maximum weight value for all six networks before applying the heat map.
- Positive weight values indicate activation, whereas negative weight values indicate repression.

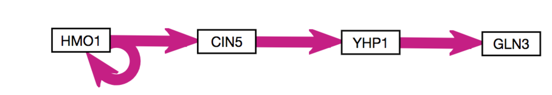
Six Related Gene Regulatory Networks Derived from the YEASTRACT Database Were Modeled with GRNmap



- GRNsight automatically generated weighted network graphs from the output produced by GRNmap.
- The absolute value of the individual weight parameters are divided by the maximum value, which distributes them between 0 and 1. Line thickness is on a linear scale with thin lines near 0 and thick lines near 1.
- Positive weights are colored magenta to indicate activation; negative weights are colored cyan to indicate repression. Weights within 5% of zero are colored gray to denote a negligible influence on the target gene.
- The ratio between the model-generated least squares error and the theoretical minimum least squares error for each network was computed. The closer the value is to one, the better GRNmap performed in modeling the GRN.

Changes in the Expression of Cin5, Hmo1, and Yhp1 Were Modeled Well; These Transcription Factors Form a Regulatory Chain

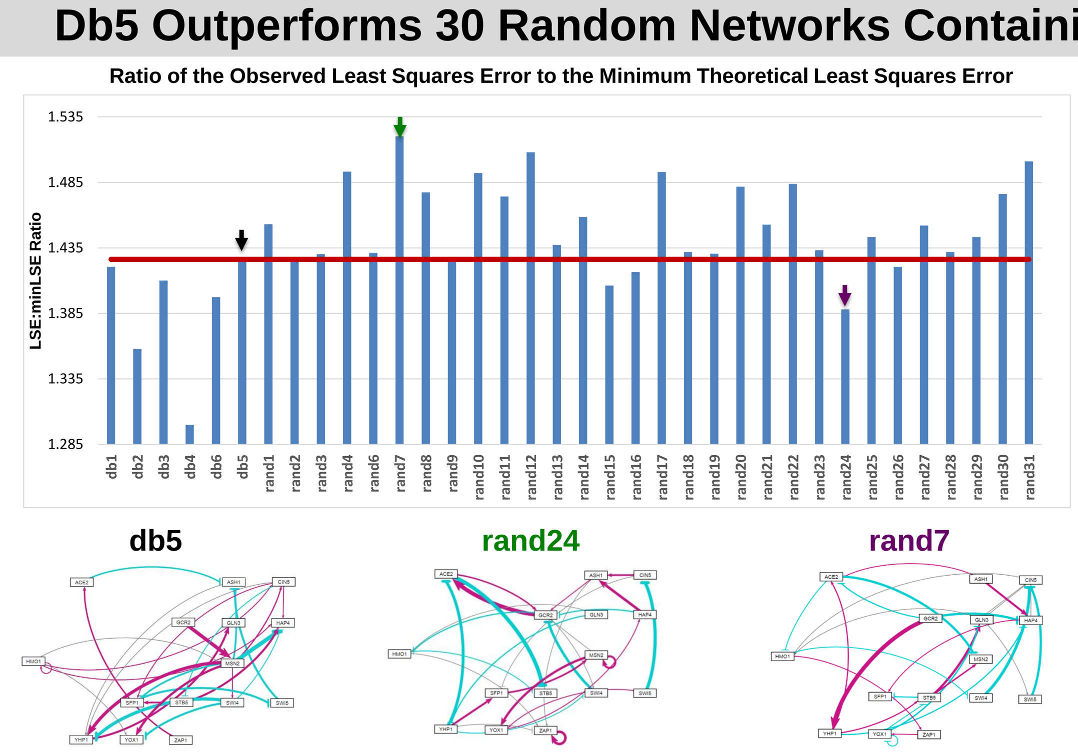




Within Strain ANOVA Benjamini and Hochberg Corrected *p* values	*HMO1*	*CIN5*	*YHP1*	*GLN3*
Wild-Type	0.0397	0.0662	0.1743	0.4189
Δ*cin5*	0.0022	—	0.8684	0.5468
Δ*gln3*	0.0025	0.0163	0.0433	0.6094
Δ*hap4*	0.4244	0.0112	0.8152	0.5189
Δ*hmo1*	—	0.4113	0.8152	0.9281
Δ*zap1*	0.0011	0.026	0.0653	0.1171

- Variability in expression dynamics is consistent with the corrected *p* values.
- Ordering the transcription factors as they appear in the motif suggests a sequential response of these genes to cold shock.

Db5 Outperforms 30 Random Networks Containing the Same Genes and Number of Edges



- Random networks were generated with the same genes as db5, the same number of edges, but randomized relationships.
- Each random network was then modeled with GRNmap.
- db5 generally provided a better fit to actual expression data. The average LSE:minLSE ratio of the random networks was higher than that of db5, with only 5 random networks exhibiting lower LSE:minLSE ratios when compared to db5 (red line, 1.4263).
- The average LSE:minLSE ratio for the six database-derived networks (1.3853) indicated that these GRNs exhibited better fits than even the best performing random network, rand24 (purple arrow, 1.3880).
- Random networks with better fits generally had some shared regulatory relationships with db5.

Multiple Regression Analyses Identified Determinants of Individual TF Model Fits

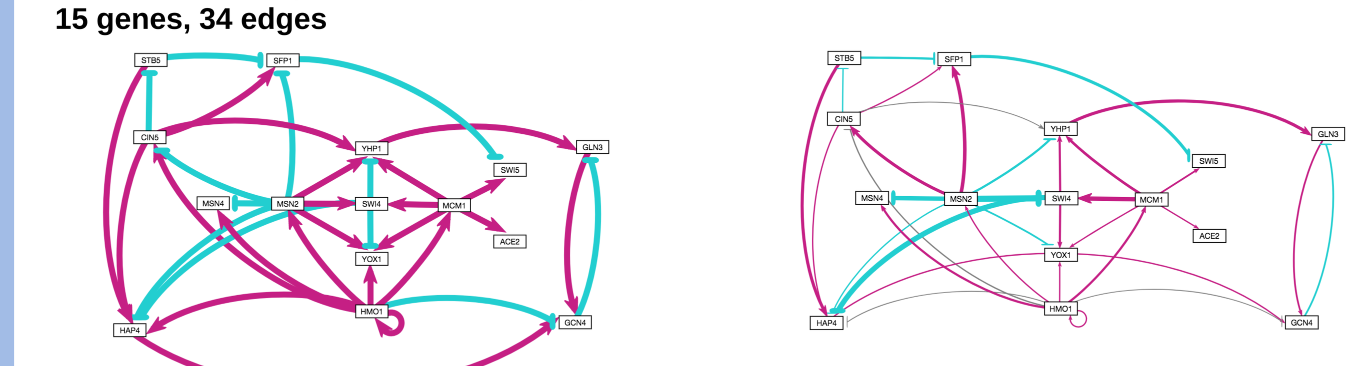
	Database-Derived Networks db1	db2	db3	db4	db5	db6	3 Best Random Networks rand15	rand16	rand24	3 Worst Random Networks rand7	rand12	rand31
ANOVA B&H p-value												
Average Expression at 15 min.	-0.507	-0.624	-0.479	-0.552	-0.528	-0.626				-0.838		-0.656
Average Expression at 30 min.												
Average Expression at 60 min.											-0.503	
Degradation Rate							-0.505		-0.611			
Optimized Production Rate	0.578			0.777		0.437						
Optimized Threshold b				0.883								
Weighted In-degree						0.617						
Weighted Out-degree							-0.627			0.433		
Eccentricity												
Closeness Centrality							-0.443					
Betweenness Centrality								-0.609		0.399		
Clustering		0.522	0.473		0.501							
Eigenvector Centrality	-0.384			-0.533		-0.328			-0.598		-0.685	-0.831
Modularity Class	—	—	—	—	—	—					0.510	
Page Ranks	—	—	—	—	—	—						0.633
Strong Component Number	—	—	—	—	—	—		0.633			-1.033	
Adjusted R^2	0.290	0.525	0.360	0.441	0.350	0.196	0.436	0.552	0.257	0.500	0.692	0.437

- Multiple regression was conducted with the mean squared error for the expression of each gene as the dependent variable.
- Predictors included model parameters and graph statistics derived from the GRNs. Stepwise regression was performed via backward elimination.
- Factors that were *not significant* predictors of MSE:minMSE ratio:
  - ANOVA Benjamini & Hochberg corrected p value
  - Average Expression at 30 minutes
  - Eccentricity
- Factors that were *consistently significant* predictors of MSE:minMSE ratio:
  - Average Expression at 15 minutes
  - Eigenvector Centrality

A Consolidated Network was Constructed from Conserved Motifs and High Eigenvector Centrality Genes

Consolidated network with predicted activation and repression relationships, 15 genes, 34 edges

Modeling results support the validity of the consolidated network



- LSE:minLSE = 1.34
- 25 out of 34 regulatory relationships predicted correctly

Acknowledgments

For their work on the GRNmap code, we would like to thank Trixie Anne M. Roque, Chukwuemeka E. Azinge, and Justin Kyle T. Torres. We thank Nicole A. Anguiano, Anindita Varshneya, Mihir Samdarshi, Edward Bachuora, Jen Shin, and Eileen Choe for their work on the GRNsight visualization software. Microarray data were collected by Cybele Arsan, Wesley Citti, Kevin Entzminger, Andrew Herman, Monica Hong, Heather King, Lauren Kubeck, Stephanie Kuelbs, Elizabeth Liu, Matthew Mejia, Kevin McGee, Kenny Rodriguez, Olivia Sakhon, Alondra Vega, and Kevin Wyllie. This work was supported by NSF-RUI Award 0921038 (K.D.D., B.G.F.) and a Kadner-Pitts Research Grant (K.D.D.).

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Availability

GRNmap software: http://kdahlquist.github.io/GRNmap/

GRNsight software: http://dondi.github.io/GRNsight/

Data: NCBI GEO Series GSE83656

Protocols: https://openwetware.org/wiki/Dahlquist:Protocols